

A Model to Test Cartilage Mediated Sound Transmission

A thesis presented in partial fulfillment of the
requirements for the degree of Bachelor of Science

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Abstract

Cartilage mediated headphones (CMH) are devices that can deliver sound via vibrations applied directly to the skin overlying cartilaginous surfaces of the outer ear. These devices typically sit outside of the ear canal, enabling users to enjoy audio without occluding their ear canals to external sound. However, shortcomings in CMH optimization have led to unresolved issues, such as sound leakage at high volumes and optimized location to interface CMH to the outer ear, discouraging wider adaption. Currently, there is no standardized method for evaluating and comparing these devices that would help advance the sound pathway.

This project aims to build a physical model of the outer ear capable to objectively simulating the acoustic transmission of sound from a transducer vibrating the pinna to the resulting sound pressure at the tympanic membrane. To achieve this, the model replicates the anatomical geometry and material properties of the outer ear including cartilage, soft tissue, and bone that compose the pinna and ear canal using a combination of PDMS-MoldMax60 composite, PDMS-Ecoflex50-BaTiO₃ composite, and Composite-X Resin respectively. The model integrates all three materials to capture the anatomical morphology and material properties of the outer ear, with the goal of creating an accessible and reproducible platform for acoustic testing and research.

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Chapter 1

Introduction

1.1 Background and Motivation

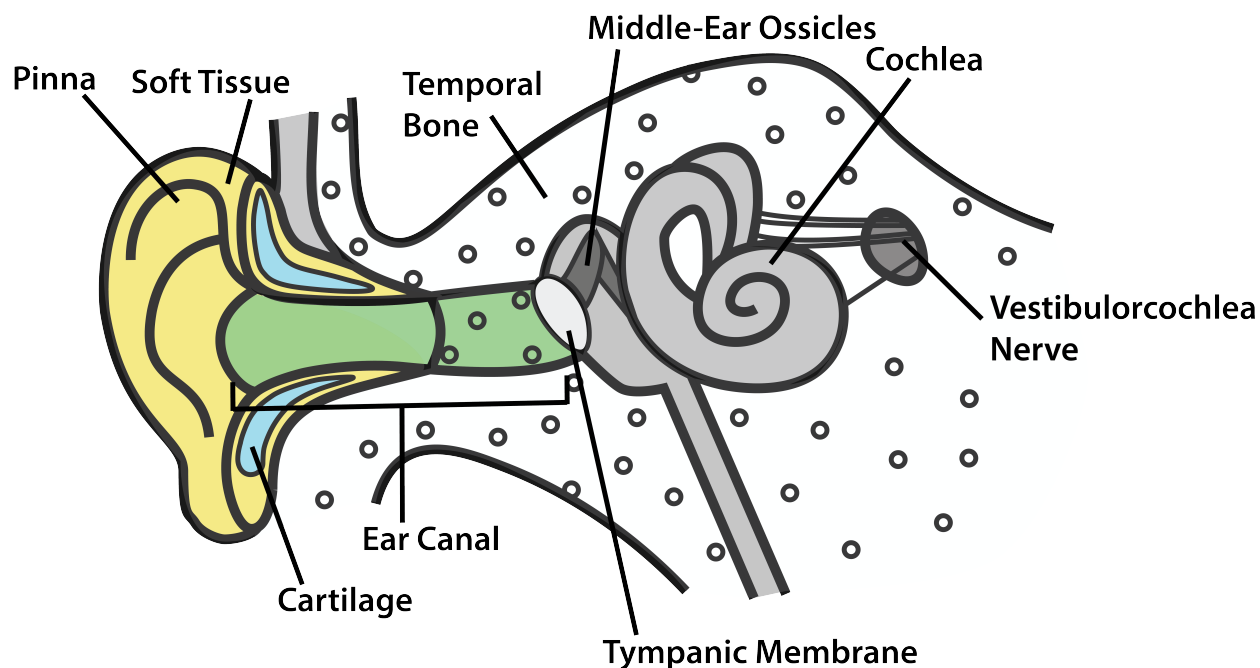


Figure 1.1: Diagram of the human ear showing the outer ear (colored) and the middle and inner ear (in gray).

Headphones are an indispensable tool in the modern world, allowing users to carve out a personal bubble in any space enabling communication, focused work, relaxation, and privacy [3]. In-the-ear headphones are a popular device that rests inside of the ear canal and transmits sound waves through the air, a process called **air conduction (AC)**. Incoming sound waves cause the tympanic membrane (ear drum) to vibrate, passing these vibrations

from the middle ear to the inner ear. These sounds are then filtered and amplified through a series of mechanical steps before they are translated into neural signals that are relayed to the brain (Figure 1.1).

However, the rising popularity of open-ear headphones, such as Shokz, reflects growing user demand for an alternative platform among users Figure 1.2. Unlike conventional headphones that transmit audio directly into the ear canal, these devices deliver sound through mechanical contact applied to the outer ear¹. Typically, these devices are mounted on cartilaginous regions around the pinna² a method we define as **cartilage-mediated sound transmission**. Strong scientific evidence supports the idea that the outer ear is the major contributor in cartilage-mediated transmission in a healthy normal ear[4][5]. A vibrating actuator applied to the pinna induces vibrations in the ear-canal cartilage, generating sound waves that propagate in the ear canal causing the tympanic membrane (eardrum) to vibrate, thus, transmitting sound to the middle and inner ear in a manner similar to conventional air conduction Figure 2.1. This open ear design enables users to experience sound from the surrounding environment while simultaneously enjoying audio from their headphone device.



Figure 1.2: Shokz OpenRun Headphones. These headphones transmit sound via vibrations applied directly on the outer ear cartilage, allowing users to enjoy audio privately, leaving the ear canal open for ambient sound.

¹The outer ear includes the visible portion of the ear, the pinna, and the ear canal up to the tympanic membrane. Its external structure, known as the pinna, is composed mainly of soft tissue and cartilage. The pinna functions to direct sound into the ear canal and towards the tympanic membrane, and like a horn, increases sound pressure from the external environment to the tympanic membrane. The peripheral half of the ear canal is surrounded by cartilage while the medial half of the ear canal is surrounded by bone.

²The visible section of the outer ear, also known as the auricle

However, current technological limitations of cartilage-mediated sound prevent wider adaptation in wearable devices. According to Shokz, the primary limitations of their headphones are as follows [6]:

- limited sound quality
- sound leakage at high volumes/quiet environments

Controlled testing of the sound transmission and contact forces would help optimize for these limitations. Several questions remain unanswered: How does the location of transducer placement and how it interacts with the pinna influence sound transmission? How would such differences in location and coupling affect the frequency response of sound trans

Previous methods for testing **Cartilage-Mediated Headphones (CMH)**³ rely on observational data from human participants. While valuable, human subject testing is hindered by its subjective nature, participant recruitment, and strict ethical regulations, making it inaccurate, time-consuming, and costly [7][8]. **Objective testing** physical data collected in a controlled manner without active subject participation offers a more consistent and efficient alternative.

The lack of a standardized testing framework continues to present an obstacle to advancement in the field. To address this gap, we propose a full-scale physical model designed specifically to test cartilage-mediated sound, which prioritizes accurate representation of the interior structure and material/acoustic properties of the outer ear.

1.2 Social Impact of CMH Advancement

Traditional in-ear headphones often rely on a sealed fit design around the ear canal to increase air-conducted sound transmission, reduce environmental noise, and preserve privacy (Figure 1.3). However, they come at the cost of occluding ambient sound, which can be dangerous. Between 2004 and 2011, 116 reported cases of death or injury of pedestrians were linked to headphone use, with nearly 30% of these incidents involving pedestrians who failed to hear warning signs. This highlights the importance of alternative devices that preserve situational awareness while still providing private audio.

³Terminology among commercial devices often varies between bone conduction and cartilage conduction, with some making conflicting claims that can be confusing. For clarity purposes, we define all instances where the transducer is placed on the cartilaginous regions of the outer ear as cartilage-mediated sound.

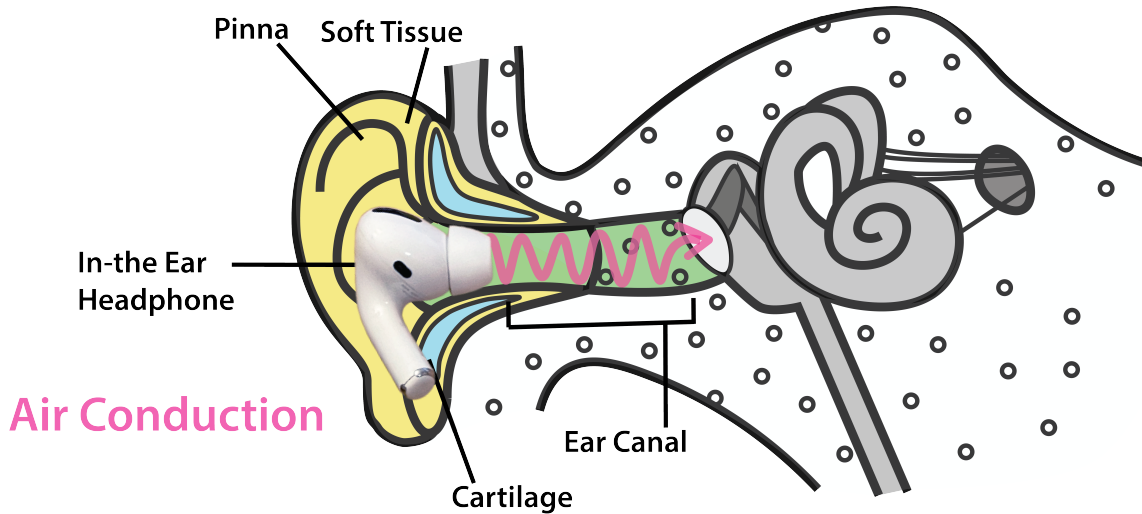


Figure 1.3: Diagram of the human ear with traditional in-ear headphones. Traditional in-ear headphones deliver sound directly into the ear canal via Air Conduction. Their sealed-fit design block the ear canal, limiting environmental awareness.

In response, many users have turned to CMH, which are growing in popularity among athletes and outdoor enthusiasts who prioritize safety by seeking headphones that allow them to remain alert to surrounding sounds [9]. The appeal of CMH, however, extends beyond recreational use. Substantial applications are becoming realized in emerging domains such as military, healthcare, and specialized niche sectors as the technology advances [10][9]. For example, CMH devices being implemented for communication in dangerous work environments such as construction sites, where workers need to stay alert to external hazards [11].

While CMH has historically been targeted towards healthy individuals, its potential for advancing assistive devices for the visual impaired are being realized [12]. According to the World Health Organization (WHO), 2.2 billion people suffer from vision impairment globally [13]. For these individuals who rely heavily on sound as an alternative sensing mechanism, a technology that can deliver additional signal input without occluding environmental awareness is especially valuable. A 2021 paper, for example, demonstrated a navigation device that implemented cartilage-mediated sound transmission to help blind individuals get around in the city [12].

As these applications expand, the economic significance of CMH is also growing. A 2023 analysis reported the market size of the technology to be 1 billion USD [9]. Experts project this number to grow up to 7 billion USD by 2033.